

Trapezoidal Grid Reconstruction for Efficient Non-Line-of-Sight Imaging

Talha Sultan, Chaoying Gu, Alex Bocchieri, Xiaochun Liu, Pavel Polynkin, and Andreas Velten

Abstract—Non-line-of-sight (NLOS) imaging reconstructs hidden scenes by analyzing indirect light transport, often using time-of-flight (ToF) measurements captured from a visible relay surface. Fast reconstruction algorithms such as Rayleigh-Sommerfeld Diffraction (RSD) exploit the Fast Fourier Transform (FFT) for efficiency but are fundamentally constrained to reconstructing on uniformly spaced Cartesian grids. Since NLOS spatial resolution degrades linearly with distance from the relay surface, standard Cartesian grids lead to inefficient oversampling in distant regions. Directly sampling the reconstruction volume with spherical coordinates offers a solution by inherently matching the resolution fall-off and producing natural perspective reconstructions where distant objects appear smaller. However, the non-uniform nature of this sampling and the complex transformations needed make direct computation significantly more demanding than with Cartesian grids. We propose a trapezoidal grid reconstruction scheme using the Scaled RSD algorithm, which integrates the Scaled Fast Fourier Transform (SFFT) with the Standard RSD. This grid, which approximates conventional spherical coordinates, retains per-voxel efficiency of Cartesian methods while requiring fewer voxels than a Cartesian grid for the same volume. Scaled RSD computes trapezoidal volumes whose max-projection visualizations are perspective-consistent, so farther scene regions occupy fewer pixels when viewed from the relay wall. We analytically derive voxel scaling strategies that efficiently match the system's depth-dependent resolution limits. This enables scalable reconstruction of large hidden volumes, balancing computational efficiency with alignment to theoretical resolution limits.

Index Terms—Computational Photography, Time-of-flight Imaging, Non-Line-of-Sight Imaging

1 INTRODUCTION

NON-LINE-OF-SIGHT (NLOS) imaging addresses the challenging problem of reconstructing objects that are hidden from direct observation by exploiting indirect optical reflections [1], [2]. Among various proposed techniques, time-of-flight (ToF)-based approaches [3], [4], [5], [6], [7] have demonstrated the ability to generate live video reconstructions of room-sized hidden scenes at 5 frames per second [8] by illuminating a relay surface with pulsed lasers and capturing photon time-of-flight measurements that are processed with a computational algorithm (Fig. 1A).

Numerous computational approaches have been proposed for NLOS reconstruction. Deep learning methods can produce high-quality results but require extensive training and may generalize poorly beyond their training domain. Iterative optimization techniques offer accuracy but are often too slow for real-time applications. In contrast, fast linear inverse methods—such as the Rayleigh-Sommerfeld Diffraction (RSD) algorithm [9], the Light Cone Transform (LCT) [10], and fk-migration [11]—leverage the Fast Fourier Transform (FFT) to achieve efficient reconstructions. While these approaches are highly optimized for speed, they are limited to reconstructing the hidden scene on a discrete,

uniformly spaced voxel grid (Fig. 1A).

However, the spatial resolution of non-line-of-sight reconstructions degrades linearly with increasing distance from the relay wall [10], [12]. This is similar to the resolution loss seen in line-of-sight systems, such as conventional cameras and LiDAR. This results in oversampling when using a Cartesian grid to reconstruct at large distances. A more efficient alternative is to sample the hidden volume along radial rays emanating from a central projection point, which better matches the spherical projection geometry of a typical line-of-sight camera. Under this view, distant regions can be represented with coarser sampling consistent with the system's depth-dependent resolution limit. This approach produces a more natural appearance in the reconstruction, with distant objects rendered smaller. However, directly computing voxels along these spherical rays is computationally expensive. Consequently, the fastest available method involves generating a high-resolution Cartesian grid, applying a depth-dependent filter, and then interpolating the results into a spherical grid.

In this work, we propose a trapezoidal grid reconstruction scheme that better approximates the spherical projection of conventional vision systems (Fig. 2). This approach maintains the per-voxel computational and memory efficiency of Cartesian grid methods while requiring fewer voxels to reconstruct a fixed volume.

1.1 Contributions

Our contributions in this paper include:

- Development of the Scaled Rayleigh-Sommerfeld Diffraction algorithm, integrating the Scaled Fast

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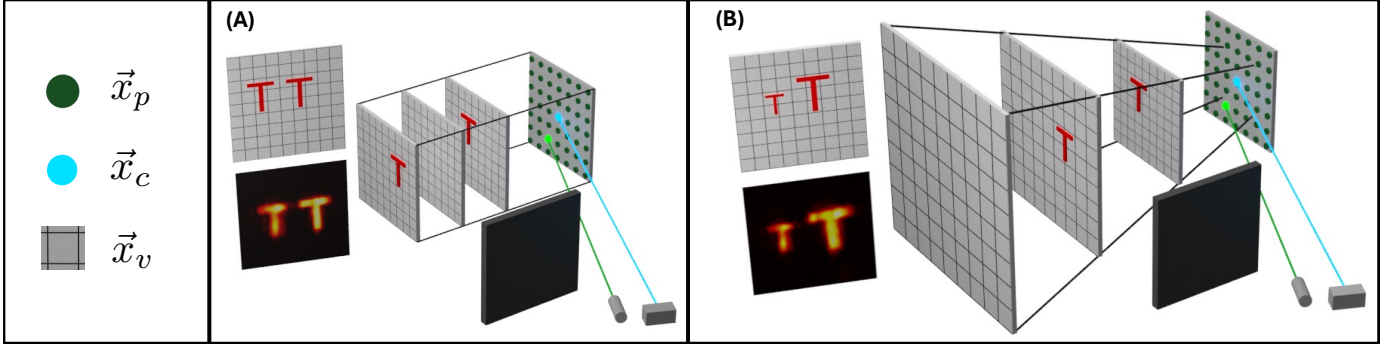


Fig. 1: **(A)** An NLOS active imaging system illuminates a relay surface with a pulsed laser and uses a ToF sensor to acquire measurements. The RSD algorithm efficiently reconstructs the hidden scene plane by plane but is limited to reconstructing a uniformly spaced Cartesian grid. A 2D image is generated from the 3D reconstruction by applying a max intensity projection along the depth axis. **(B)** Our Scaled RSD algorithm addresses the limitations of Standard RSD by reconstructing larger scenes within the same memory footprint. This is achieved by increasing the voxel size as a function of depth.

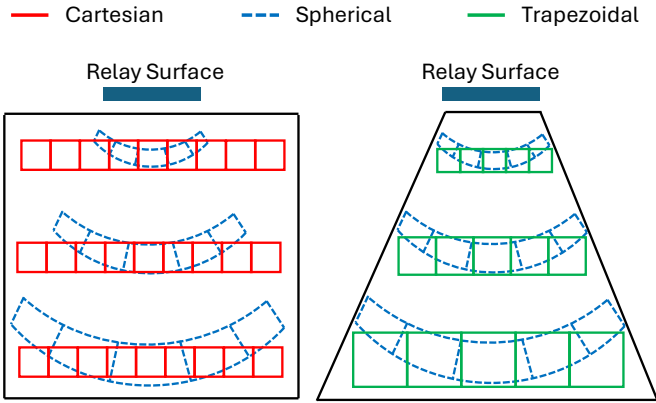


Fig. 2: **Voxel grid sampling schemes.** The spherical sampling grid (blue) correctly accounts for resolution loss with depth. The Cartesian grid (left) oversamples at larger distances from the relay wall, while the trapezoidal grid (right) more closely approximates spherical sampling and reduces oversampling.

Fourier Transform with Standard RSD to compute trapezoidal-grid reconstructions. These volumes support perspective-consistent max-projection visualizations, in which farther scene regions occupy fewer pixels when projected along depth. (Fig. 1B)

- Analytical demonstration of optimal voxel scaling strategies that match output sampling to the depth-dependent resolution limit, enabling larger reconstructed volumes under a fixed voxel budget.

Representative code and scripts supporting this work are publicly available [13].

2 RELATED WORK

2.1 FFT-based NLOS Reconstruction

Most practical NLOS systems rely on efficient solvers that formulate reconstruction as a convolution and accelerate the computation using the Fast Fourier Transform (FFT). The Light Cone Transform (LCT) [10] and f-k migration algorithms [11] handle confocal data - where the illumination and detection points are co-located on the relay

wall and jointly raster-scanned on a planar relay surface. Both methods reconstruct the full 3D volume using 3D FFTs—LCT via deconvolution with a light-cone kernel, and f-k migration through wave-based backpropagation and Stolt interpolation. In contrast, the Rayleigh–Sommerfeld Diffraction (RSD) algorithm [9] supports both confocal and non-confocal acquisition, where the illumination and detection points are scanned independently. It reconstructs the scene one depth slice at a time using 2D FFTs. This slice-wise structure improves memory efficiency and enables depth-dependent voxel scaling, which better mimics perspective projection. Such adaptive scaling is difficult to achieve with global 3D FFT-based approaches like LCT and f-k migration, which reconstruct the entire voxel grid at the same resolution.

2.2 Phasor Field Framework

The phasor field framework provides a wave-based interpretation of non-line-of-sight (NLOS) imaging, recasting it as a coherent line-of-sight (LOS) imaging problem from the perspective of the relay wall [12], [14], [15], [16], [17], [18]. In this formulation, light transport is modeled as the propagation of a virtual wavefront, and reconstruction is performed via diffraction-based backpropagation—most notably through the Rayleigh–Sommerfeld Diffraction (RSD) algorithm [9], [19]. This framework draws strong analogies between NLOS imaging and established coherent imaging systems such as holography, ultrasound, and phased arrays. It has been used to derive resolution limits [12], [14], [20], define sampling criteria [21], provide intuition about the missing cone problem in NLOS imaging [22], and extract complex light transport within the hidden scene [23].

2.3 Resolution Fall-Off and Depth-Adaptive Sampling

Phasor field analysis reveals that the lateral resolution of RSD-based reconstructions follows the Rayleigh criterion and decays linearly with distance from the relay wall. This resolution fall-off closely mirrors what is observed in physical LOS cameras and motivates the view of RSD as a computational camera. However, whereas physical cameras preserve angular resolution by sampling along rays [24],

FFT-based NLOS algorithms typically reconstruct the scene on a Cartesian voxel grid with uniform resolution. This mismatch leads to oversampling at greater depths, wasting memory and computation by exceeding the system's resolution limit.

Time-domain methods such as ellipsoidal filtered back-projection (FBP) offer the flexibility to sample voxels on non-Cartesian grids—including spherical and trapezoidal layouts—that naturally account for resolution loss [3], [6], [25]. However, their high computational cost limits their use in large-scale or real-time settings. To address this inefficiency, our work integrates a trapezoidal sampling strategy into the RSD framework, allowing voxel size to adapt with depth. This approach retains FFT-based efficiency while better approximating the sampling characteristics of physical cameras.

2.4 Large Hidden Scenes

While NLOS imaging has been demonstrated at large distances between the imaging system and the relay surface [26], reconstructing large hidden volumes remains challenging due to resolution degradation and reduced signal-to-noise ratio (SNR) at greater depths. Nam et al. [8] use phasor field analysis to design a depth-dependent frame averaging strategy, showing that the SNR remains approximately constant until the target becomes smaller than the system's resolution limit. This averaging strategy is complementary to our Scaled RSD approach, which reconstructs plane by plane; integrating both methods could effectively address SNR degradation and resolution fall-off in future large-scale reconstructions.

3 BACKGROUND: PHASOR-FIELD IMAGING

The methodologies presented in this paper leverage the phasor field framework for non-line-of-sight (NLOS) imaging. This approach conceptually transforms the acquired NLOS measurements into data resembling a complex wavefront captured by a line-of-sight (LOS) coherent imaging system. By employing diffraction integrals derived from wave optics, this virtual wavefront can be computationally propagated backward into the concealed volume, thereby producing three-dimensional reconstructions of hidden structures. The subsequent discussion reviews the fast Rayleigh-Sommerfeld Diffraction (RSD) algorithm detailed in [27], which processes measurements in the temporal frequency domain. A thorough understanding of this algorithm is crucial, as our contributions directly extend its capabilities.

3.1 Standard RSD Algorithm

The RSD algorithm is able to generate reconstructions of the hidden scene from both confocal and non-confocal measurements and can be accelerated using the FFT [8], [27]. The main idea of the RSD is that it encodes the temporal shifts in the measurement as phasors in the Fourier domain, so time shifts in the reconstruction algorithm can be computed using complex addition. Furthermore, Fourier-based convolution can be used to describe plane-to-plane diffraction and accelerate the calculation using the Fast Fourier Transform (FFT). Consequently, the RSD algorithm

maintains accuracy only when both the planar relay wall and each reconstruction plane are sampled uniformly.

The raw data consists of transient measurements, denoted $H(\vec{x}_p, \vec{x}_c, t)$, captured at various illumination positions $\vec{x}_p$ and detection positions $\vec{x}_c$ on the relay surface over time t . Assuming the light transport within the system is linear and time-invariant for each spatial channel, H can be interpreted as the impulse response function of the hidden scene. Consequently, the scene's virtual response, $\mathcal{P}(\vec{x}_c, t)$, to an arbitrary virtual illumination pattern, $\mathcal{P}(\vec{x}_p, t)$, can be determined by temporally convolving the impulse response with the illumination function:

$$\mathcal{P}(\vec{x}_c, t) = \mathcal{P}(\vec{x}_p, t) * H(\vec{x}_p, \vec{x}_c, t) \quad (1)$$

Invoking Helmholtz reciprocity [28], [29], the roles of illumination and detection can be interchanged. In this view, each detection point becomes a virtual phasor-field source at $\vec{x}_p$, while the original illumination positions form a virtual aperture at $\vec{x}_c$. This perspective is particularly useful in single-pixel NLOS systems, where a fixed detector records responses from a raster-scanned laser [9].

Transforming this virtual response into the temporal frequency domain yields $\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega)$, where each frequency component ω corresponds to a phasor. This complex value encapsulates the phase accumulated by a hypothetical monochromatic wavefront as it travels from an illumination point $\vec{x}_p$, interacts with the hidden scene, and returns to a detection point $\vec{x}_c$. The RSD integral is then employed to computationally backpropagate each such wavefront from the relay surface into the hidden volume, ultimately forming the 3D reconstruction, $I(\vec{x}_v)$, evaluated at voxel coordinates $\vec{x}_v$:

$$I(\vec{x}_v) = \left| \int \underbrace{\frac{e^{-j\frac{\omega}{c}|\vec{x}_p - \vec{x}_v|}}{|\vec{x}_p - \vec{x}_v|}}_{\mathcal{P}_{\mathcal{F}}(\vec{x}_p)} \underbrace{\left[\int \mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega) \frac{e^{-j\frac{\omega}{c}|\vec{x}_v - \vec{x}_c|}}{|\vec{x}_v - \vec{x}_c|} d\vec{x}_c \right]}_{R_{\vec{x}_v}(\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega))} d\omega d\vec{x}_p \right|^2 \quad (2)$$

Here, $\mathcal{P}_{\mathcal{F}}(\vec{x}_p)$ acts as a phase mask dependent on the illumination position, while $R_{\vec{x}_v}(\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega))$ represents a phase mask associated with the detection positions. The evaluation of the term $R_{\vec{x}_v}$ can be significantly expedited using FFT-based convolution, provided the detection points $\vec{x}_c$ constitute a uniformly sampled grid on a planar surface:

$$R_{\vec{x}_v}(\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega)) = \mathcal{P}_{\mathcal{F}}(x_c, y_c, 0, \omega) *_{2D} G(x_c, y_c, z_v, \omega) \quad (3)$$

The term $G(x_c, y_c, z_v, \omega) = \frac{e^{i\frac{\omega}{c}\sqrt{x_c^2 + y_c^2 + z_v^2}}}{\sqrt{x_c^2 + y_c^2 + z_v^2}}$ serves as the propagation kernel. Notably, this kernel is translationally invariant with respect to (x_c, y_c) for any given depth plane z_v . This property allows for the simultaneous reconstruction of all voxels $\vec{x}_v$ within a specific plane, enabling the 3D volume to be constructed in a plane-by-plane manner.

For computations on a uniform grid, any discrete location (x_c, y_c) can be expressed as $(m\Delta x, n\Delta y)$, where

$(\Delta x, \Delta y)$ are the spatial sampling intervals and (m, n) are integer indices within ranges like $m \in [-M/2, M/2 - 1]$ and $n \in [-N/2, N/2 - 1]$. The plane-to-plane propagation via convolution can then be efficiently implemented using FFTs:

$$\begin{aligned} \mathcal{P}_{\mathcal{F}}(m\Delta x, n\Delta y, z_v, \omega) = \\ \text{IFFT} \left\{ \text{FFT} \left\{ \mathcal{P}_{\mathcal{F}}(m\Delta x, n\Delta y, 0, \omega) \right\} \right. \\ \left. \cdot \text{FFT} \left\{ G(m\Delta x, n\Delta y, z_v, \omega) \right\} \right\} \end{aligned} \quad (4)$$

3.2 Confocal Reconstruction

The Standard RSD algorithm can be adapted for confocal NLOS measurements, where the illumination and detection points coincide (i.e., $\vec{x}_p = \vec{x}_c = \vec{x}_s$). In this case, the light travels along the same path twice, resulting in a round-trip propagation model. The RSD integral can be simplified using a spatial Dirac delta function and reduces to:

$$I(\vec{x}_v) = \int \mathcal{P}_{\mathcal{F}}(\vec{x}_s, \omega) e^{-2j\frac{\omega}{c}|\vec{x}_v - \vec{x}_s|} d\omega d\vec{x}_s, \quad (5)$$

To account for the doubled path length, the propagation kernel is evaluated at twice the frequency:

$$G'(x_s, y_s, z_v, 2\omega) = \frac{e^{i\frac{2\omega}{c}\sqrt{x_s^2 + y_s^2 + z_v^2}}}{\sqrt{x_s^2 + y_s^2 + z_v^2}}. \quad (6)$$

This formulation retains compatibility with the FFT-based convolution used in the non-confocal case, enabling efficient depth-by-depth reconstruction. Further details can be found in [9].

3.3 Computational Complexity

Following established notation [9], the computational complexity of this approach is analyzed. Assuming the reconstruction volume is sampled at the same rate as the relay surface (potentially after zero-padding to extend the lateral range) and forms a cube with side length N samples, the algorithm requires performing N sets of 2D FFTs (across the depth dimension) for a fixed number of frequency components ω . This results in a computational complexity of $O(N^3 \log N)$ for processing data corresponding to a single illumination position $\vec{x}_p$. To distinguish this baseline method from the modified version introduced later, we will refer to it as the Standard RSD. For comparison, alternative methods like Filtered Backprojection (FBP) typically solve the RSD integral directly but with a higher complexity of $O(N^5)$ [30].

3.4 Phasor Field Sampling and Resolution

Although the virtual phasor field illumination $\mathcal{P}_{\mathcal{F}}(\vec{x}_p, \omega)$ can, in principle, be arbitrary, it is commonly defined as a bandpass filter in the frequency domain, centered at ω_c :

$$\mathcal{P}_{\mathcal{F}}(\vec{x}_p, \omega) = \delta(\vec{x} - \vec{x}_p) e^{-\frac{(\omega - \omega_c)^2}{2\sigma^2}} \quad (7)$$

The central frequency ω_c corresponds to a central wavelength $\lambda_c = 2\pi c/\omega_c$, where c is light speed. The filter's bandwidth is controlled by σ , often related to λ_c (e.g.,

$\sigma = c/(5\lambda_c)$). To prevent aliasing according to the Nyquist criterion, the relay wall should ideally be spatially sampled at intervals of $\lambda_s = \lambda_c/2$ or finer [30]. Under these conditions, the achievable lateral resolution, Δx , for a reconstructed point source within the hidden volume at depth z from a relay surface of diameter D , has been established as [12], [14], [17], [20]:

$$\Delta x = \frac{1.22\lambda_c z}{D} \quad (8)$$

This expression reveals that lateral resolution degrades linearly with increasing depth z and central wavelength λ_c . This inherent characteristic implies that the Standard RSD algorithm, which employs a fixed spatial sampling rate across all depths, effectively oversamples the reconstruction grid at larger distances where resolution is naturally coarser.

4 METHODS: SCALED RSD

We introduce the Scaled RSD as an extension of the original RSD that computes plane-to-plane diffraction. Using an FFT to accelerate the reconstruction means that the spatial sampling rate on the planar relay surface, x_c , determines the output sampling rate of each reconstructed plane. Our goal is to computationally change the sampling rate of the output plane without changing the sampling rate at the input plane, since the latter is governed by hardware and may be difficult to modify physically. Furthermore, this modification should not change the underlying computational complexity of the reconstruction algorithm.

4.1 Math of Scaled RSD

The relay-wall measurements remain sampled on the fixed hardware grid, $x_c = m\Delta x$ and $y_c = n\Delta y$. Scaled RSD changes only the coordinates at which each reconstructed output plane is evaluated. Thus, $\mathcal{P}_{\mathcal{F}}(\alpha x_c, \beta y_c, z_v, \omega)$ denotes the propagated field evaluated on a scaled output grid, not a resampling of the relay-wall measurements.

With this convention, we rewrite Eq. 3 as

$$\begin{aligned} \mathcal{P}_{\mathcal{F}}^{\alpha, \beta}(x_c, y_c, z_v, \omega) &\equiv \mathcal{P}_{\mathcal{F}}(\alpha x_c, \beta y_c, z_v, \omega) \\ &= \mathcal{P}_{\mathcal{F}}(x_c, y_c, 0, \omega) *_{\alpha, \beta} G(\alpha x_c, \beta y_c, z_v, \omega), \end{aligned} \quad (9)$$

where $*_{\alpha, \beta}$ denotes the SFFT-accelerated propagation from fixed input samples to scaled output coordinates.

The propagation kernel is evaluated at the scaled output coordinates using its definition:

$$G(\alpha x_c, \beta y_c, z_v, \omega) = \frac{e^{i\frac{\omega}{c}\sqrt{(\alpha x_c)^2 + (\beta y_c)^2 + z_v^2}}}{\sqrt{(\alpha x_c)^2 + (\beta y_c)^2 + z_v^2}}. \quad (10)$$

To evaluate the propagated field on the scaled output grid without resampling the measured input field, we utilize the Scaled Fast Fourier Transform (SFFT) [31], [32]. This allows non-integer scaling of the Fourier-domain representation while keeping the input wavefront uniformly sampled. Computing a 2D SFFT requires the input signal to be uniformly sampled:

$$\begin{aligned} \mathcal{P}_{\mathcal{F}}(x_c, y_c, 0, \omega) &= \mathcal{P}_{\mathcal{F}}(m\Delta x, n\Delta y, 0, \omega), \\ (m, n) &\in \mathcal{I}_{M, N}. \end{aligned} \quad (11)$$

where we define

$$I_{\mathcal{L}} := \{\ell \in \mathbb{Z} : -\mathcal{L}/2 \leq \ell < \mathcal{L}/2\} \quad (12)$$

so $\mathcal{I}_{M,N}$ is a 2D grid. Defining the 2D DFT kernel as:

$$U[m', n'] \triangleq \sum_{m=-M/2}^{M/2-1} \sum_{n=-N/2}^{N/2-1} u[m, n] \cdot k_M^{-m \cdot m'} k_N^{-n \cdot n'} \quad (13)$$

where

$$k_{\mathcal{L}} = \exp \left[-j \frac{2\pi}{\mathcal{L}} \right]. \quad (14)$$

In Supplement Section S1.1, we derive that the SFFT, i.e. the Fourier-domain propagation result evaluated on the scaled output grid $(\alpha x_c, \beta y_c)$, can be computed using three additional FFTs:

$$\begin{aligned} \text{FFT} \left\{ \mathcal{P}_{\mathcal{F}}^{\alpha, \beta}(m\Delta x, n\Delta y, z_v, \omega) \right\} &= k_M^{-\frac{\alpha}{2} m'^2} \cdot k_N^{-\frac{\beta}{2} n'^2} \\ \text{IFFT} \left\{ \text{FFT} \left\{ \mathcal{P}'_{\mathcal{F}}(m\Delta x, n\Delta y, 0, \omega) \right\} \cdot \text{FFT} \left\{ k_M^{\frac{\alpha}{2} m^2} \cdot k_N^{\frac{\beta}{2} n^2} \right\} \right\}. \end{aligned} \quad (15)$$

where

$$\mathcal{P}'_{\mathcal{F}}(m\Delta x, n\Delta y, 0, \omega) = \mathcal{P}_{\mathcal{F}}(m\Delta x, n\Delta y, 0, \omega) \cdot k_M^{-\frac{\alpha}{2} m^2} \cdot k_N^{-\frac{\beta}{2} n^2}. \quad (16)$$

Computing three additional FFTs does not increase the asymptotic computational complexity. This allows Scaled RSD to evaluate a depth-scaled output grid without first reconstructing a larger zero-padded Cartesian volume. In the next sections, we show that this reduces the number of stored and evaluated samples for matched physical coverage.

4.2 Scaling Factors

The goal of scaling is to increase the reconstructed cross-sectional area linearly with depth without changing the number of stored samples per plane. Thus, we define the output physical side length at depth z explicitly by:

$$x_{\text{out}}(z) = x_{\text{in}} + k_x z, \quad y_{\text{out}}(z) = y_{\text{in}} + k_y z \quad (17)$$

yielding depth-dependent scaling factors:

$$\alpha(z) = \frac{x_{\text{in}}}{x_{\text{in}} + k_x z}, \quad \beta(z) = \frac{y_{\text{in}}}{y_{\text{in}} + k_y z} \quad (18)$$

where $x_{\text{in}}, y_{\text{in}}$ indicate the initial lateral side lengths (Fig. 3), and z_{in} and z_{out} are the initial and final depths from the relay surface. As shown in Fig. 1 and Fig. 3, the shape of the reconstruction volume changes from a *cuboid* to a *pyramidal frustum*.

In this framework, when $0 < \alpha(z), \beta(z) < 1$, the physical cross-sectional area increases as depth z increases. For simplicity, we assume $x_{\text{out}} = y_{\text{out}}$ for any z so $k_x = k_y = k$. Under this isotropic scaling assumption, we show in Supplement Section S1.2 that the volume of this specific frustum is defined by:

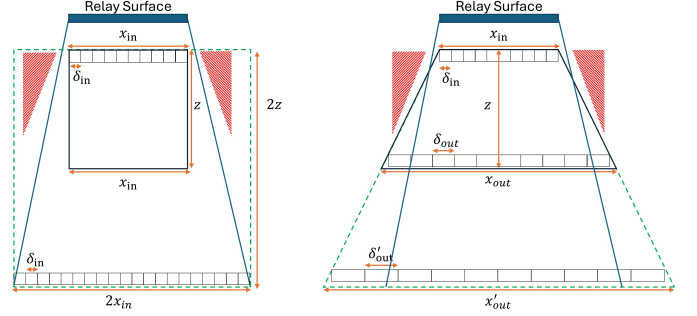


Fig. 3: Showing the top view of the reconstruction volume when reconstructing with the RSD algorithm (left) and the Scaled RSD algorithm (right). The dotted green lines demonstrate the increase in volume as we increase z , when the lateral field of view (FOV) is fixed. The RSD algorithm requires zero padding and increasing the side length, while the Scaled RSD increases the lateral FOV without increasing the number of pixels.

$$V_F = \left(H x_{\text{in}}^2 + H^2 x_{\text{in}} k + H^3 \left(\frac{k^2}{3} \right) \right) \quad (19)$$

where H is the height of this frustum.

As shown in Fig. 3, the output pixel side length, $\delta_{\text{out}}(z)$, increases proportionally with z (Fig. 3), and is given by:

$$\alpha(z) = \frac{\delta_{\text{in}}}{\delta_{\text{out}}(z)} = \frac{x_{\text{in}}}{x_{\text{out}}(z)} \quad (20)$$

Increasing the output voxel size compensates for the resolution loss while reducing the number of samples needed for reconstruction. This raises a key question: what is the maximum voxel size that can be used for reconstruction without losing information? The lateral resolution from Eq. 8 provides a basis for deriving upper and lower bounds respectively for $\delta_{\text{out}}(z)$ and $\alpha(z)$:

$$\delta_{\text{out}}(z) \leq \frac{\Delta x}{2} = \frac{1.22 \lambda_c z}{2D} \quad (21)$$

$$\alpha(z) \geq \frac{2\delta_{\text{in}}}{\Delta x} = 2 \frac{\delta_{\text{in}}}{\Delta x} \quad (22)$$

4.2.1 Choosing the scaling rate k

To avoid under-sampling resolvable lateral structure at larger depths, we enforce a resolution-safe upper bound on the scaling rate,

$$k \leq k_{\text{max}} \triangleq \frac{x_{\text{in}}}{z_{\text{max}}} \left(\frac{1.22 \lambda_c z_{\text{max}}}{2D \delta_{\text{in}}} - 1 \right). \quad (23)$$

We choose $k \leq k_{\text{max}}$ in all experiments. The derivation of (23) is provided in Supplement Sec. S1.3, and Supplement Sec. S3.1 empirically validates the equivalent normalized sampling condition $\delta_{\text{out}}/\delta_{\text{limit}} \leq 1$.

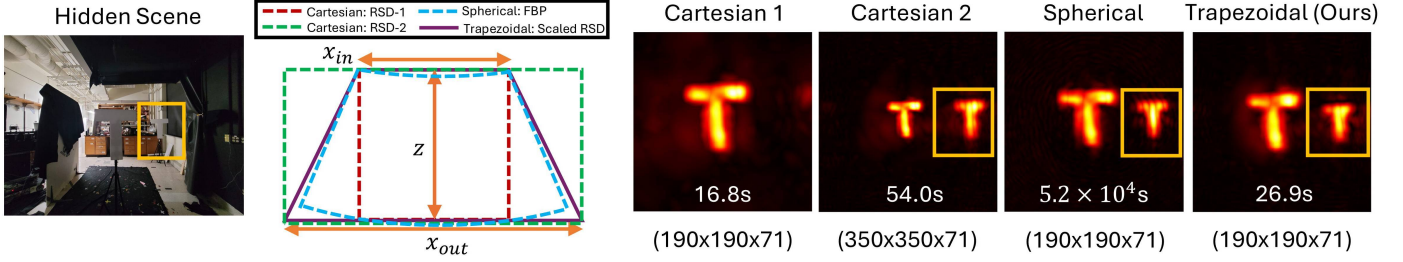


Fig. 4: Column 1 shows the hidden scene as viewed from the relay surface. Column 2 provides a top view of the reconstruction grids, illustrating that under a fixed voxel budget the Cartesian grid covers a smaller lateral area than the spherical and trapezoidal grids. Columns 3–6 show 2D images generated by applying a max projection along depth to reconstructions obtained using a Cartesian grid with $(190 \times 190 \times 71)$ voxels (Column 3), a zero-padded Cartesian grid with $(350 \times 350 \times 71)$ voxels (Column 4), a spherical grid with $(190 \times 190 \times 71)$ voxels (Column 5), and a trapezoidal grid with $(190 \times 190 \times 71)$ voxels (Column 6). Both the spherical and trapezoidal grids recover the rightmost letter “T” under the fixed voxel budget, whereas the Cartesian grid does not. The spherical reconstruction is computed using time-domain FBP and therefore requires substantially longer runtime. Average reconstruction time is shown in white text at the bottom center of each image, and the black text below indicates the reconstruction size ($X \times Y \times Z$).

4.3 Complexity Analysis

The computational complexity reported in the literature assumes the reconstruction volume is a cube. For the Standard RSD, this volume is equal to $V_C = z^3$ since $z = x_{in} = y_{in}$. The number of samples used in memory is equal to $Z^3 = (z/\delta)^3$. As shown by the solid black lines in Fig. 3, we note that the Scaled RSD algorithm uses the same number of samples to reconstruct a larger volume and the difference is governed by

$$\Delta V = V_F - V_C = \left(k + \frac{k^2}{3}\right) z^3 \quad (24)$$

when $x_{in} = z$. The two volumes are identical when k is zero and $\alpha(z) = 1$. As a concrete example, if $k = 2$, $z = 4$ m, then $V_C = 64$ m³, $V_F = 277$ m³, $\Delta V = 213$ m³, and the percentage increase in volume, $\Delta V/V_C$, is 333%.

We analyze the computational complexity for a fixed field of view (FOV) as a function of depth, as indicated by the two blue lines in Fig. 3, for a virtual camera placed behind the relay wall. The existing implementation of the Standard RSD [9] requires zero padding to increase the side length to match the field of view (Fig. 3 left), before performing three 2D FFTs ($O(Z^2 \log Z)$ steps) to reconstruct each plane. Therefore, the number of output samples/voxels scale by $O(Z^3)$ while the computation scales by $O(Z^3 \log Z)$. The Scaled RSD, by contrast, adjusts the output sample size (δ_{out} in Fig. 3) at each depth to match the required FOV by computing 3 additional FFTs. This approach adds a constant number of samples per depth, and computes a total of six 2D FFTs for each depth for $N^2 = (x/\delta_{in})^2$ output samples, matching the number of input samples on the relay surface (Fig. 13). Therefore, the number of output samples scale by $O(Z)$ while the computational complexity scales by $O(ZN^2 \log N)$.

We note that the Standard RSD algorithm can be easily modified to output a frustum by changing the zero padding as a function of depth. This adjusted implementation can reduce the number of samples used in memory, but is still suboptimal since the pixel size remains fixed and therefore, does not account for resolution loss as z increases. Additionally, the Standard RSD can achieve the same complexity

as the Scaled RSD by removing the FOV requirement and reconstructing the same size plane at each depth but this means we reconstruct a significantly smaller volume.

5 RESULTS

We evaluate whether trapezoidal sampling provides a more efficient reconstruction grid for NLOS imaging than a uniform Cartesian grid, while retaining the main geometric benefits of spherical sampling. Because NLOS lateral resolution decreases with depth, a uniform Cartesian grid allocates voxels inefficiently in distant regions. Our analysis therefore focuses on three questions: (1) whether trapezoidal grids enable larger reconstructed volumes under a fixed voxel budget, (2) whether they better approximate spherical-grid behavior than Cartesian grids, and (3) whether they preserve the depth-dependent appearance expected under perspective projection.

We apply Scaled RSD to measurements collected with the experimental setup shown in Fig. S1. In all experiments, we use the same scaling factor α along both lateral dimensions. Because the exposure time per spatial grid position is approximately 3 ms, we average measurements from three to five illumination positions $\vec{x}_p$ to improve signal-to-noise ratio (SNR). The runtime reported in each figure denotes the average reconstruction time per illumination position. For a fixed number of voxels in memory, Scaled RSD requires approximately 1.5 $\times$ more computation time than Standard RSD due to three additional FFT operations. The scaling rule used throughout these experiments is empirically validated in Supplement Sec. S3.1.

We compare Scaled RSD against Standard RSD and FBP baselines. Within each comparison, all methods use the same phasor-field wave packet parameters. All runtimes are CPU-only and were obtained on the same workstation without GPU/CUDA acceleration. Standard RSD and Scaled RSD are implemented in MATLAB, while FBP is evaluated using a CPU C++ implementation parallelized with pthreads. For FBP, we report thread-normalized CPU time, computed as the measured wall-clock time multiplied by the number of pthreads. These timings are intended to illustrate the relative computational cost of the methods

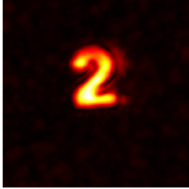
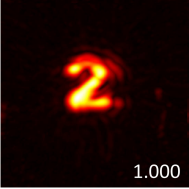
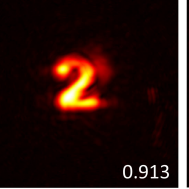
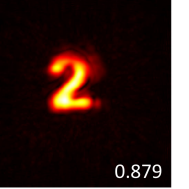
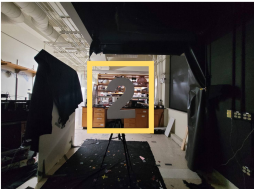
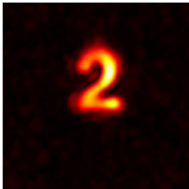
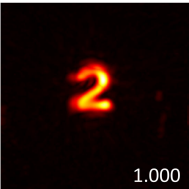
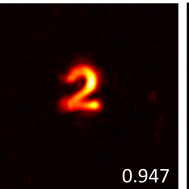
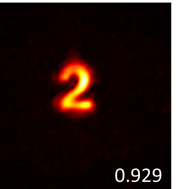
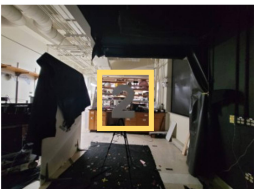
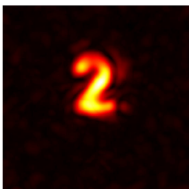
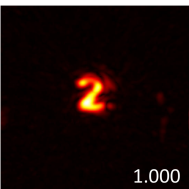
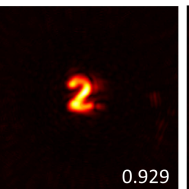
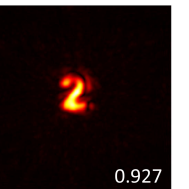
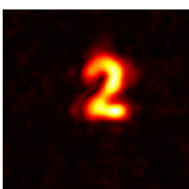
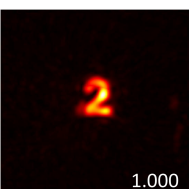
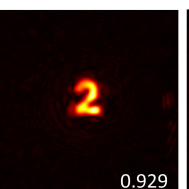
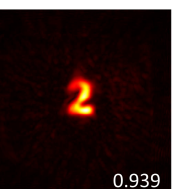
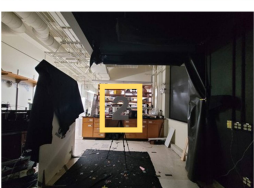
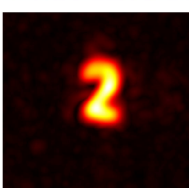
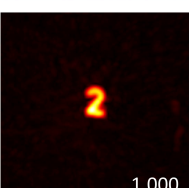
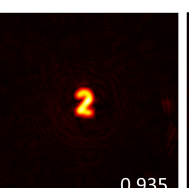
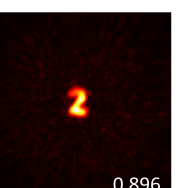
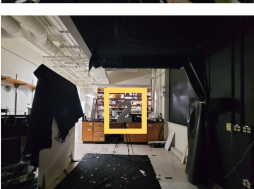
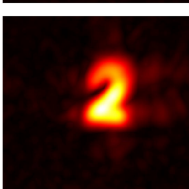
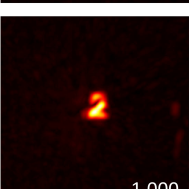
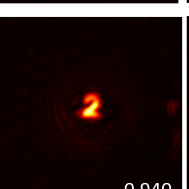
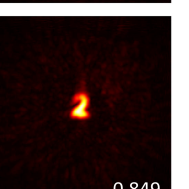
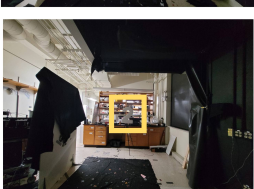
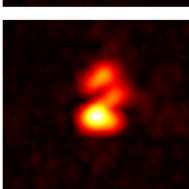
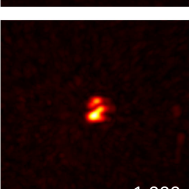
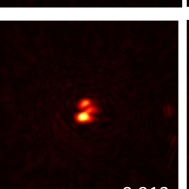
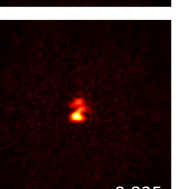
Depth	Hidden Scene	RSD Cartesian	FBP Spherical	FBP Trapezoidal	Scaled RSD (Ours) Trapezoidal
1.20 m			 1.000	 0.913	 0.879
1.48 m			 1.000	 0.947	 0.929
1.80 m			 1.000	 0.929	 0.927
2.13 m			 1.000	 0.929	 0.939
2.84 m			 1.000	 0.935	 0.896
3.32 m			 1.000	 0.940	 0.849
3.65 m			 1.000	 0.913	 0.825
Average Runtime		16.7 s	50,251.50 s	50,448.16 s	25.6 s
Volume Reconstructed		9.68 m ³	39.60 m ³	41.92 m ³	41.92 m ³

Fig. 5: We place the digit “2” at different depths within the hidden scene and reconstruct each scene using a fixed voxel budget of $190 \times 190 \times 68$. Column 1 lists the average depth of the digit for each row. Column 2 shows a reference photograph of the hidden scene, with a yellow box marking the digit location. Columns 3–6 show reconstructions obtained using Standard RSD on a Cartesian grid, FBP on a spherical grid, FBP on a trapezoidal grid, and Scaled RSD on a trapezoidal grid, respectively. Because spherical FBP, trapezoidal FBP, and Scaled RSD reconstruct on depth-scaled grids, the digit occupies fewer pixels as its distance from the relay surface increases, consistent with perspective projection from the relay-wall viewpoint. In contrast, Standard RSD reconstructs on a uniform Cartesian grid and therefore produces an orthographic-like visualization. The spherical FBP reconstruction is used as the reference for image metrics. For the FBP trapezoidal and Scaled RSD columns, the bottom-right number reports SSIM after maximum-intensity normalization, orientation matching, and 2D correlation alignment. The final rows report the average runtime and reconstructed volume for each method.

under a controlled CPU-only comparison, rather than to compare fully optimized implementations.

We first compare Cartesian, spherical, and trapezoidal reconstruction grids on a controlled scene, and then evaluate the effect of grid choice on depth-dependent resolution and perspective-consistent appearance. Finally, we test the method on large-scale simulations and on public datasets with complex hidden scenes.

5.1 Comparison of Reconstruction Grids

We begin by comparing Cartesian, spherical, and trapezoidal reconstruction grids on a controlled scene. Figure 4 highlights the first practical consequence of this design choice: volumetric coverage under a fixed voxel budget. With the voxel count fixed to $(190 \times 190 \times 71)$, the Cartesian grid reconstructs a smaller lateral area and fails to capture the letter “T” on the right side of the scene (Column 3). Both the spherical grid (Column 5) and the trapezoidal grid (Column 6) recover the desired geometric extent using the same voxel budget. However, the spherical reconstruction is implemented here using time-domain filtered backprojection (FBP) and therefore requires substantially more computation, taking 5.2×10^4 s. A Cartesian reconstruction can recover similar lateral coverage only by increasing the grid size through zero-padding (Column 4), which raises the reconstruction volume to $(350 \times 350 \times 71)$ voxels and increases runtime to 54.0 s. In contrast, the trapezoidal grid achieves comparable coverage using $(190 \times 190 \times 71)$ voxels and 26.9 s.

To quantify how well the trapezoidal grid approximates the spherical grid, we compute SSIM between the spherical-grid FBP reconstruction and the trapezoidal reconstruction. The full-image SSIM is 0.80. Most of the remaining discrepancy is due to low-level background artifacts rather than differences in the reconstructed object. After masking background regions by removing pixels with intensity below 10% of the image maximum, the SSIM increases to 0.97, indicating close agreement in the reconstructed hidden “T” letters. Trapezoidal sampling therefore provides a practical middle ground: it approaches the geometric coverage of spherical sampling while retaining FFT-based efficiency and requiring fewer voxels than the zero-padded Cartesian alternative.

TABLE 1: Quantitative analysis on the Fig. 4 two-target “T” scene under matched depth-scaled lateral coverage. For Cartesian methods, we report the enclosing Cartesian volume required to cover the same lateral extent as the depth-scaled reconstructions.

	Cartesian RSD matched coverage	Scaled RSD trapezoidal	FBP spherical	Cartesian RSD + spherical resampling
Runtime (s)	54.0	26.9	5.2×10^4	55.1
Stored voxels	$350 \times 350 \times 71$	$190 \times 190 \times 71$	$190 \times 190 \times 71$	$350 \times 350 \times 71$
Output storage (MiB)	66.4	19.6	19.6	≥ 66.4
Volume stored/evaluated (m^3)	34.3	23.6	24.4	34.3
Complexity	$O(N^3 \log N)$	$O(N^3 \log N)$	$O(N^5)$	$O(N^3 \log N)$
Storage scaling	$O(Z^3)$	$O(Z)$	$O(Z)$	$O(Z^3)$

5.2 NLOS Depth Resolution

We use NLOS measurements of the digit “2” placed at different depths within the hidden scene to empirically illustrate the depth dependence of NLOS lateral resolution. Figure 5

summarizes the results. Column 1 lists the average depth of the hidden object, and Column 2 shows a reference image captured by a smartphone from the relay-wall viewpoint. Columns 3–6 show reconstructions obtained using Standard RSD on a Cartesian grid, FBP on a spherical grid, FBP on a trapezoidal grid, and Scaled RSD on a trapezoidal grid, respectively.

As the digit moves farther from the relay surface, the Standard RSD reconstruction shows the expected loss of lateral detail, consistent with the depth-dependent resolution limit of the system. In contrast, spherical FBP, trapezoidal FBP, and Scaled RSD reconstruct on depth-scaled grids, so the reconstructed digit appears smaller at larger depths, consistent with the perspective visible in the smartphone image. The spherical-grid FBP reconstruction is used as the reference for the image metrics reported in the FBP trapezoidal and Scaled RSD columns. This comparison separates the effect of output geometry from the reconstruction operator: FBP on the trapezoidal grid tests how closely the trapezoidal grid approximates spherical sampling, while Scaled RSD tests the FFT-based scaled operator on the same trapezoidal output grid.

Full depth-resolved SSIM, masked SSIM, PSNR, and masked PSNR values are reported in Supplementary Tables S1 and S2. We include masked metrics because image-wide SSIM and PSNR become increasingly sensitive to low-intensity background and reconstruction artifacts as the perspective-projected digit occupies fewer pixels at larger depths.

Scaled RSD does not recover detail beyond the physical resolution limit; rather, it increases voxel size with depth so that output sampling remains better matched to the system’s decreasing lateral resolution. Although this experiment is limited by the size of our laboratory space, it illustrates the central motivation for Scaled RSD: improving sampling efficiency at larger depths while remaining consistent with the observed resolution falloff.

5.3 Scaled RSD on Large Simulations

To study performance beyond the depth range available in our laboratory, we simulate a large hidden scene consisting of three planar targets, each of size $1 \text{ m} \times 1 \text{ m}$. The targets are placed at depths of 3.33 m, 6.66 m, and 10 m, with lateral positions of 0 m, 3.7 m, and -5.7 m, respectively. We reconstruct this scene using Standard RSD on a Cartesian grid and Scaled RSD on a trapezoidal grid. For the same reconstruction coverage, Standard RSD requires $742 \times 742 \times 426$ voxels and 5661.5 s, whereas Scaled RSD requires only $298 \times 298 \times 426$ voxels and 1772.6 s. Thus, in large-scale scenes, Scaled RSD retains the desired field of view while significantly lowering both voxel count and runtime by adapting the voxel spacing to the depth-dependent resolution limit of the NLOS system.

5.4 Perspective-Consistent Max-Projection

To visualize each reconstructed 3D volume as a 2D image, we display the voxel with maximum intensity along the depth axis. When applied to the depth-scaled trapezoidal volume produced by Scaled RSD, this max-projection yields a perspective-consistent visualization: scene regions farther

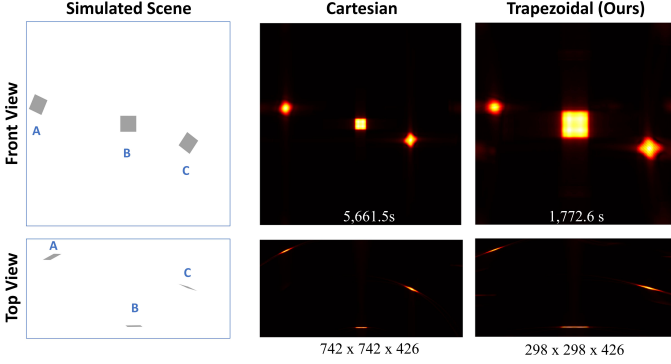


Fig. 6: Large-scale simulated scene. Left: front and top views of the simulated hidden scene, containing three planar $1\text{m} \times 1\text{m}$ targets at different depths and lateral positions. Middle: Standard RSD reconstruction on a Cartesian grid. Right: Scaled RSD reconstruction on a trapezoidal grid. Top and bottom rows show front-view and top-view maximum-intensity projections, respectively. While both methods reconstruct the same large scene extent, Scaled RSD achieves this coverage using fewer voxels and lower runtime, demonstrating the benefit of depth-adaptive sampling for large hidden volumes.

from the relay wall occupy fewer output pixels. In contrast, applying the same max-projection to a uniformly sampled Cartesian volume produces an orthographic-like visualization, since lateral voxel spacing is fixed across depth.

Fig. 7 shows reference photographs of the hidden scenes from the relay-wall viewpoint in Column 1. Column 2 shows the Standard RSD reconstruction on a Cartesian grid, while Column 3 shows the Scaled RSD reconstruction on a depth-scaled trapezoidal grid. In Row 1, the farther letter “T” occupies fewer pixels in the Scaled RSD visualization, consistent with perspective projection. Row 2 shows letters T, W, and U with different physical sizes placed at different depths so that they subtend similar apparent sizes from the relay-wall viewpoint. The Scaled RSD max-projection better preserves this depth-dependent apparent size, whereas the Cartesian RSD max-projection displays the letters closer to their physical sizes on the uniform grid.

5.5 Complex Scenes with Confocal and Non-Confocal Datasets

In this section, we evaluate Scaled RSD on more complex scenes beyond our controlled letter targets. We use publicly available confocal and non-confocal NLOS datasets that include more challenging structure, such as clutter, non-planar geometry, specular surfaces, and, when available, objects distributed across multiple hidden depths [11], [30]. Because most publicly available NLOS datasets contain objects concentrated at a single depth, such examples remain limited. Even so, these results provide a more realistic test of the method than our controlled planar scenes. Overall, Scaled RSD recovers scene structure across both acquisition schemes while using a depth-adaptive reconstruction grid.

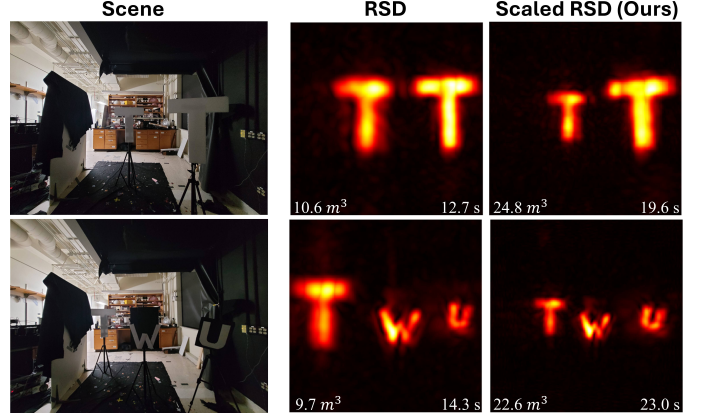


Fig. 7: Column 1 shows the reference photographs of the hidden scene. Columns 2 and 3 display the Standard RSD reconstruction and the Scaled RSD reconstruction, respectively, after applying a max filter operation. Row 1: Two letter T’s of the same size are shown, with the T in the middle appearing smaller in the Scaled RSD reconstruction due to its greater depth. Row 2: Letters T, W, and U of different sizes are arranged at varying depths so that they appear the same size when viewed from the relay wall. The numbers at the bottom left and bottom right of each image indicate the reconstructed volume and the reconstruction time, respectively.

6 DISCUSSION

6.1 Trade-offs Between Cartesian, Spherical, and Trapezoidal Grids

Table 1 summarizes the main trade-offs between Cartesian, spherical, Cartesian reconstruction followed by spherical interpolation, and trapezoidal sampling. Cartesian grids remain attractive because they are directly compatible with FFT-based reconstruction, but they oversample distant regions of the hidden volume. Spherical sampling better matches the depth-dependent resolution falloff and yields the most natural geometry, but direct reconstruction on a spherical grid is computationally expensive. Trapezoidal sampling provides a practical compromise: it captures the main geometric benefit of spherical sampling by increasing voxel size with depth, while preserving FFT-based efficiency.

A further option is to reconstruct with Standard RSD on a Cartesian grid and then interpolate the result to a spherical grid. While this produces a spherical output, it still requires reconstructing and storing the larger Cartesian volume first, followed by an additional interpolation step. As a result, it increases runtime and does not provide the same memory savings as direct trapezoidal reconstruction.

A non-uniform FFT (NUFFT) could also be used to evaluate reconstructions on spherical or more general non-Cartesian grids. However, such an approach would require nonuniform evaluation or interpolation for each depth plane and temporal frequency, adding practical overhead relative to standard FFT-based operations. In contrast, Scaled RSD uses the Scaled FFT for the specific linear coordinate scaling needed by the trapezoidal grid, preserving the slice-wise FFT structure of RSD while directly evaluating depth-scaled output coordinates.

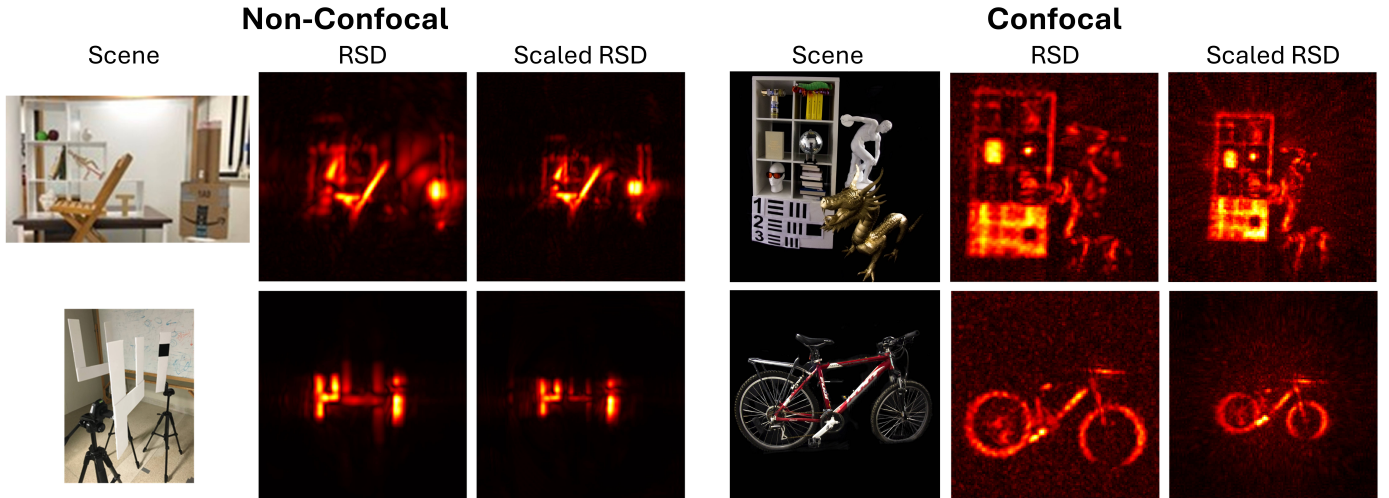


Fig. 8: We evaluate Standard RSD and Scaled RSD on publicly available NLOS datasets acquired using both non-confocal and confocal scanning schemes. Columns 1–3 show two non-confocal scenes (one per row), including a reference image of the hidden scene (Column 1) and the corresponding max-projection reconstructions obtained with Standard RSD (Column 2) and Scaled RSD (Column 3). Columns 4–6 show two confocal scenes (one per row), including a reference image (Column 4) and the corresponding reconstructions from Standard RSD (Column 5) and Scaled RSD (Column 6). These scenes are more complex than our controlled letter targets and include clutter, non-planar geometry, objects at multiple depths when available, and varied surface reflectance, including specular components. Across both acquisition schemes, Scaled RSD recovers scene structure comparable to Standard RSD while using a depth-adaptive reconstruction grid.

TABLE 2: Quantitative implementation details for the public NLOS datasets shown in Fig. 8. Each column corresponds to one reconstructed dataset in Fig. 8. We report grid size, stored voxel count, CPU runtime, output storage, and reconstructed physical volume for Standard RSD and Scaled RSD. Runtimes are CPU-only and are reported per reconstruction.

	Non-confocal		Confocal	
	Scene 1 Office Scene [30]	Scene 2 Multiple letters [30]	Scene 3 Cluttered shelf [11]	Scene 4 Bicycle [11]
Standard RSD grid	$300 \times 300 \times 44$	$200 \times 200 \times 19$	$128 \times 128 \times 64$	$128 \times 128 \times 39$
Scaled RSD grid	$300 \times 300 \times 44$	$200 \times 200 \times 19$	$128 \times 128 \times 64$	$128 \times 128 \times 39$
Standard RSD runtime (s)	9.11	1.23	3.37	2.12
Scaled RSD runtime (s)	18.07	2.45	7.64	4.19
Standard RSD storage (MiB)	30.21	5.80	8.00	4.88
Scaled RSD storage (MiB)	30.21	5.80	8.00	4.88
Standard RSD volume (m^3)	15.84	3.04	4.10	4.99
Scaled RSD volume (m^3)	36.77	7.05	9.42	11.49

6.2 Scope

Scaled RSD does not recover spatial resolution beyond the physical limits of the NLOS system and does not improve the photon signal-to-noise ratio. Its role is instead to allocate voxels more efficiently by matching output sampling to the system’s depth-dependent resolution. Its main benefit is therefore improved volumetric coverage and reduced memory usage under a fixed voxel budget, rather than improved intrinsic reconstruction quality. In low-SNR regimes or at very large depths, complementary strategies such as longer integration, depth-dependent averaging, or denoising remain necessary. Unlike fast FFT-based methods such as LCT and f-k migration, Scaled RSD supports both confocal and non-confocal datasets. In addition, LCT and f-k migration reconstruct the full hidden volume using global 3D FFTs on a fixed Cartesian grid, which increases memory usage and does not naturally support the depth-adaptive sampling

considered here.

6.3 Compatibility with Complementary Methods

Scaled RSD is complementary to other methods that improve the practicality of large-scale NLOS imaging. It can be combined with depth-dependent averaging strategies that address SNR degradation at larger distances [8], as well as methods that reduce the memory and computational footprint of the propagation kernel by exploiting its radial structure [33], [34]. Recent phase-based methods such as Zero-Phase Phasor Fields are complementary in a different dimension: ZPPF improves depth precision and z-axis sampling efficiency through phase-based refinement [35], whereas Scaled RSD reduces redundant sampling in the lateral x-y dimensions through depth-adaptive voxel scaling. Combining both could enable efficient large-volume reconstruction with improved depth localization.

7 CONCLUSION

We have presented the Scaled Rayleigh-Sommerfeld Diffraction (Scaled RSD) algorithm, an extension of Standard RSD for efficient depth-adaptive NLOS reconstruction. By integrating a Scaled Fast Fourier Transform, Scaled RSD evaluates each depth plane on a trapezoidal grid whose voxel size increases with distance from the relay wall. This parameterization matches the depth-dependent resolution falloff of NLOS imaging, reduces redundant voxel sampling, improves memory efficiency, and enables reconstruction of larger hidden volumes while preserving the slice-wise FFT structure of RSD. The resulting volumes also support perspective-consistent max-projection visualizations, in which farther scene regions occupy fewer output pixels.

Future work will explore combining Scaled RSD with complementary strategies for large-scale NLOS imaging, including phase-based depth refinement, depth-dependent averaging, NUFFT-based reconstruction on more general non-Cartesian grids, and extensions of depth-adaptive sampling to other FFT-based NLOS methods.

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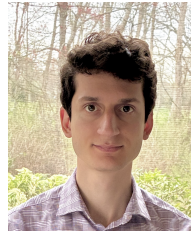
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Trapezoidal Grid Reconstruction for Efficient Non-Line-of-Sight Imaging

Supplementary Material

In Section S1, we provide the mathematical derivations referenced in the main paper. Section S2 details the hardware setup for NLOS imaging used to collect the experimental datasets. In Section S3, we present additional results.

S1 DERIVATIONS

S1.1 Scaled FFT Derivation

Starting with the DFT equation:

$$U[m', n'] \triangleq \sum_{m=-M/2}^{M/2-1} \sum_{n=-N/2}^{N/2-1} u[m, n] \cdot k_M^{-m \cdot m'} k_N^{-n \cdot n'} \quad (S1)$$

where

$$k_L = \exp \left[-j \frac{2\pi}{L} \right] \quad (S2)$$

For the rest of the derivation, we use (m, n) to represent the **integer** coordinates in the input space with $\delta_{\text{in}} = \delta$ sampling interval, and (m', n') to represent the **integer** coordinates in the output space with $\delta_{\text{in}} = \delta$ sampling rate. Note that (m, n, m', n') are all variables that range from $-N/2$ to $N/2 - 1$.

For a given frequency, we simplify the input phasor field $\mathcal{P}_{\mathcal{F}}(x_c, y_c, 0, \omega)$ to a discrete 2D representation $u[m, n]$, and the goal is to compute its Fourier transform on a scaled grid, i.e., $U[m', n']$. Use the definition of DFT, evaluating the output at the scaled coordinates introduces an extra α factor in the exponent term:

$$\begin{aligned} U_{\alpha}[m', n'] &\triangleq \mathcal{F} \{ \mathcal{P}_{\mathcal{F}}(\alpha x_c, \alpha y_c, 0, \omega) \} \\ &= \sum_{m=-N/2}^{N/2-1} \sum_{n=-N/2}^{N/2-1} u[m, n] \cdot \exp \left[\left(-j \frac{2\pi}{N} \right) \cdot -\alpha (m \cdot m' + n \cdot n') \right]. \end{aligned} \quad (S3)$$

$U_{\alpha}[m', n']$ is essentially a squeezed and resampled version of $U[m', n']$, but we stay with the integer indices notation using m' and n' because all the signals are supposed to be discrete. In the derivation that follows, we demonstrate that explicit zero-padding is not required to achieve the desired resampling of $U[m', n']$. Instead, the process can be reformulated as a set of additional FFTs, thereby accelerating the computation. We here start from the 1D case and then extend into 2D later.

$$U_{\alpha}[n'] = \sum_{n=-N/2}^{N/2-1} u[n] \cdot k_N^{-\alpha \cdot n \cdot n'} \quad \text{where} \quad k_N = \exp \left[-j \frac{2\pi}{N} \right] \quad (S4)$$

since $\alpha \cdot n \cdot n' = \alpha/2 \cdot (n^2 + n'^2 - (n' - n)^2)$, Eq. S3 can be rewritten as:

$$\begin{aligned} U_{\alpha}[n'] &= \sum_{n=-N/2}^{N/2-1} u[n] \cdot k_N^{\frac{\alpha}{2}(n'-n)^2} \cdot k_N^{-\frac{\alpha}{2}n^2} \cdot k_N^{-\frac{\alpha}{2}n'^2} \\ &= k_N^{-\frac{\alpha}{2}n'^2} \sum_{n=-N/2}^{N/2-1} \underbrace{u[n] \cdot k_N^{-\frac{\alpha}{2}n^2}}_{u'[n]} \cdot \underbrace{k_N^{\frac{\alpha}{2}(n'-n)^2}}_{k'_N[n'-n]} \\ &= k_N^{-\frac{\alpha}{2}n'^2} \sum_{n=-N/2}^{N/2-1} u'[n] \cdot k'_N[n'-n] \end{aligned} \quad (S5)$$

After defining $u'[n] = u[n] \cdot k_N^{-\frac{\alpha}{2}n^2}$ and $k'_N[n] = k_N^{\frac{\alpha}{2}n^2}$, Eq. S5 can be treated as a convolution and apply FFT to calculate.

$$U_{\alpha}[n'] = k_N^{-\frac{\alpha}{2}n'^2} \cdot \text{IFFT} \{ \text{FFT} \{ u'[n] \} \cdot \text{FFT} \{ k'_N[n] \} \} \quad (S6)$$

S1.2 Volume of Frustum

For a pyramidal frustum, the side length varies linearly with z so that:

$$x_{\text{out}}(z) = x_{\text{in}} + k_x z, \quad \alpha(z) = \frac{x_{\text{in}}}{x_{\text{in}} + k_x z} \quad (S7)$$

$$y_{\text{out}}(z) = y_{\text{in}} + k_y z, \quad \beta(z) = \frac{y_{\text{in}}}{y_{\text{in}} + k_y z} \quad (S8)$$

$$z_{\text{out}} = z_{\text{in}} + z \quad (S9)$$

Then we can derive the volume by integrating cross-sectional areas along z :

$$\begin{aligned}
V_F &= \int_{z_{\text{in}}}^{z_{\text{out}}} A(z) dz \\
&= \int_{z_{\text{in}}}^{z_{\text{out}}} (x_{\text{in}} + \alpha z) (y_{\text{in}} + k_y z) dz \\
&= \int_{z_{\text{in}}}^{z_{\text{out}}} (x_{\text{in}} y_{\text{in}} + k_y x_{\text{in}} z + k_x y_{\text{in}} z + (k_x k_y z^2)) dz \\
&= \left| \left(z (x_{\text{in}} y_{\text{in}}) + \frac{z^2}{2} (k_y x_{\text{in}} + k_x y_{\text{in}}) + \frac{z^3}{3} (k_x k_y) \right) \right|_{z_{\text{in}}}^{z_{\text{out}}}
\end{aligned} \tag{S10}$$

Assuming that each cross-section of the frustum is a square, then $x_{\text{in}} = y_{\text{in}}$ and $k_x = k_y = k$, we get:

$$V_F = \left| \left(z x_{\text{in}}^2 + z^2 x_{\text{in}} k + z^3 \left(\frac{k^2}{3} \right) \right) \right|_{z_{\text{in}}}^{z_{\text{out}}} \tag{S11}$$

Letting $z_{\text{in}} = 0$, $z_{\text{out}} = H$, we recover Eq. 24:

$$V_F = \left(H x_{\text{in}}^2 + H^2 x_{\text{in}} k + H^3 \left(\frac{k^2}{3} \right) \right).$$

where H is the height of our frustum.

S1.3 Resolution-Safe Sampling Bound

Under isotropic scaling, the lateral field of view increases with depth as

$$x_{\text{out}}(z) = x_{\text{in}} + k z, \tag{S12}$$

which implies a depth-dependent lateral voxel spacing

$$\delta_{\text{out}}(z) = \delta_{\text{in}} \left(1 + \frac{k z}{x_{\text{in}}} \right). \tag{S13}$$

To avoid under-sampling resolvable structure, the output voxel spacing must satisfy

$$\delta_{\text{out}}(z) \leq \frac{\Delta x(z)}{2}, \tag{S14}$$

where $\Delta x(z)$ is the depth-dependent lateral resolution. Using the diffraction-limited form

$$\Delta x(z) = \frac{1.22 \lambda_c z}{D}, \tag{S15}$$

we define the corresponding resolution limit

$$\delta_{\text{limit}}(z) \triangleq \frac{\Delta x(z)}{2}. \tag{S16}$$

Enforcing (S14) at the farthest depth $z = z_{\text{max}}$ gives

$$\delta_{\text{in}} \left(1 + \frac{k z_{\text{max}}}{x_{\text{in}}} \right) \leq \delta_{\text{limit}}(z_{\text{max}}), \tag{S17}$$

which yields the resolution-safe upper bound

$$k \leq k_{\text{max}} = \frac{x_{\text{in}}}{z_{\text{max}}} \left(\frac{\delta_{\text{limit}}(z_{\text{max}})}{\delta_{\text{in}}} - 1 \right). \tag{S18}$$

This condition can be interpreted as a depth-dependent Nyquist criterion for the scaled voxel grid. In the experiments below, we report the normalized ratio $\delta_{\text{out}}/\delta_{\text{limit}}$ to indicate whether the voxel spacing remains within the resolution-safe range.

S2 HARDWARE

In this work, we build an active NLOS imaging system with a PM-1.03-25TM pulsed laser from Polar Laser Laboratories as the illumination source, and a customized gated 16 x 16 pixel Single Photon Avalanche Diode (SPAD) array, developed at Politecnico di Milano [36], as the detector (Column 1, Fig. S1). The laser is coupled with a frequency doubler to generate 515 nm pulses at an average rep rate of 5 MHz and average power of 375 mW. We sequentially illuminate individual locations on a 1.9 m x 1.9 m area on a relay wall using a two-mirror Thorlabs galvanometer (Thorlabs GVS012). Furthermore, we use an Edmund Optics 6X Manual Zoom Video lens to focus our 16x16 array to a 50 cm x 32.5 cm (Width x Height) area on the relay wall. The overall temporal resolution of our system is characterized by a Full-Width at Half Maximum (FWHM) of approximately 65 ps.

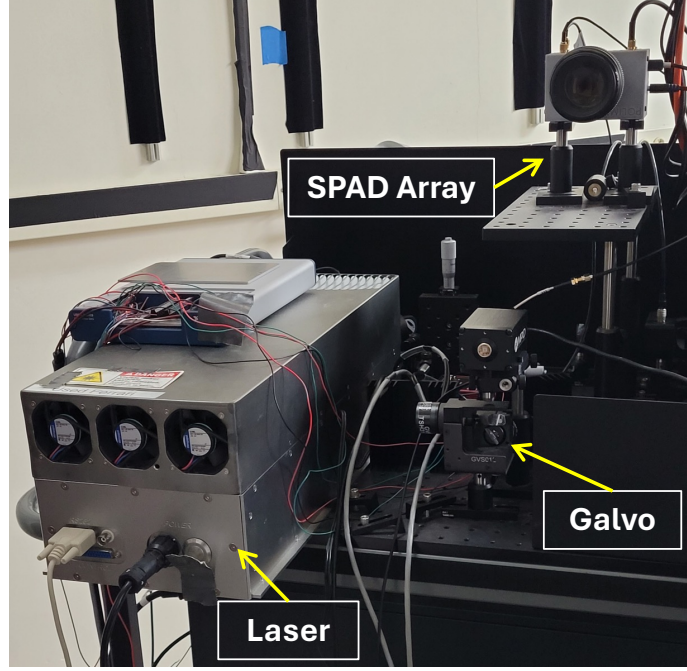


Fig. S1: An image of our NLOS imaging system consisting of a pulsed laser, galvanometer, and SPAD array.

S3 SUPPLEMENTAL RESULTS

S3.1 Empirical Validation of the Resolution-Safe Bound

Figure S2 shows reconstructions obtained for increasing values of the normalized sampling ratio $\delta_{\text{out}}/\delta_{\text{limit}}$, where δ_{limit} is defined in (S16) and the corresponding resolution-safe condition is given by (S14). All other parameters are held fixed. For ratios below unity, the reconstructions remain stable and preserve the hidden object structure. As the ratio approaches unity, mild degradation becomes visible. Once the bound is exceeded, ringing and structured aliasing artifacts appear and rapidly dominate the reconstruction. This behavior is consistent with Fourier-domain aliasing: when the scaled voxel spacing exceeds the resolution-safe limit, resolvable spatial frequencies are no longer sampled adequately and fold back into the reconstructed image.

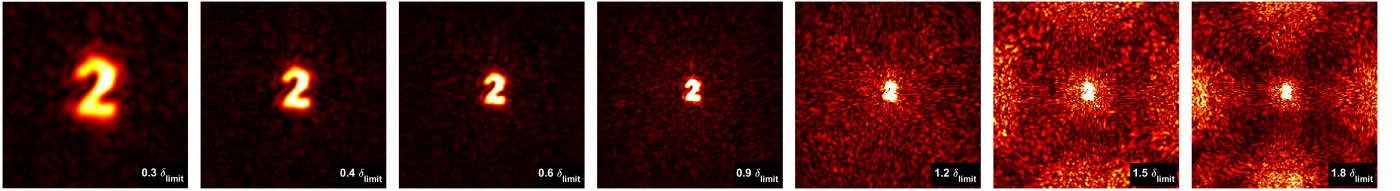


Fig. S2: Effect of exceeding the resolution-safe sampling limit. Reconstructions for increasing values of the normalized ratio $\delta_{\text{out}}/\delta_{\text{limit}}$, with all other parameters fixed. Below the limit, the hidden object remains stable and interpretable. Near the limit, mild degradation appears. Beyond the limit, ringing and structured aliasing artifacts rapidly degrade the reconstruction.

S3.2 Scaled RSD on Non-Uniform Acquisition schemes

The Scaled RSD algorithm requires the measurement to be collected with a uniformly sampled spatial grid on the relay surface, which typically requires additional calibration. One strategy that has worked for other RSD based methods [37], is to sample the relay surface with a non-uniform grid, indicated by $\vec{x}'_c$, and interpolate to a regular grid ($\vec{x}_c$):

$$\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega) = \phi(\mathcal{P}_{\mathcal{F}}(\vec{x}'_c, \omega)) \quad (\text{S19})$$

Then, we can proceed with the Scaled RSD as usual. In Fig. S3, we demonstrate that this strategy works well on experimental data from [19]. This dataset was collected with a uniform grid — to recreate a non-uniform acquisition grid, we randomly subsample the relay surface while simultaneously discarding 90% of the grid positions (Column 1). Then we reconstruct with the total grid (Row 1) and the non-uniform sparse grid (Row 2) using the Standard RSD (Columns 2 and 4) and the Scaled RSD (Columns 3 and 5). Our results show that the Scaled RSD can generate reconstructions using non-uniform sparse grids without substantial loss in image quality.

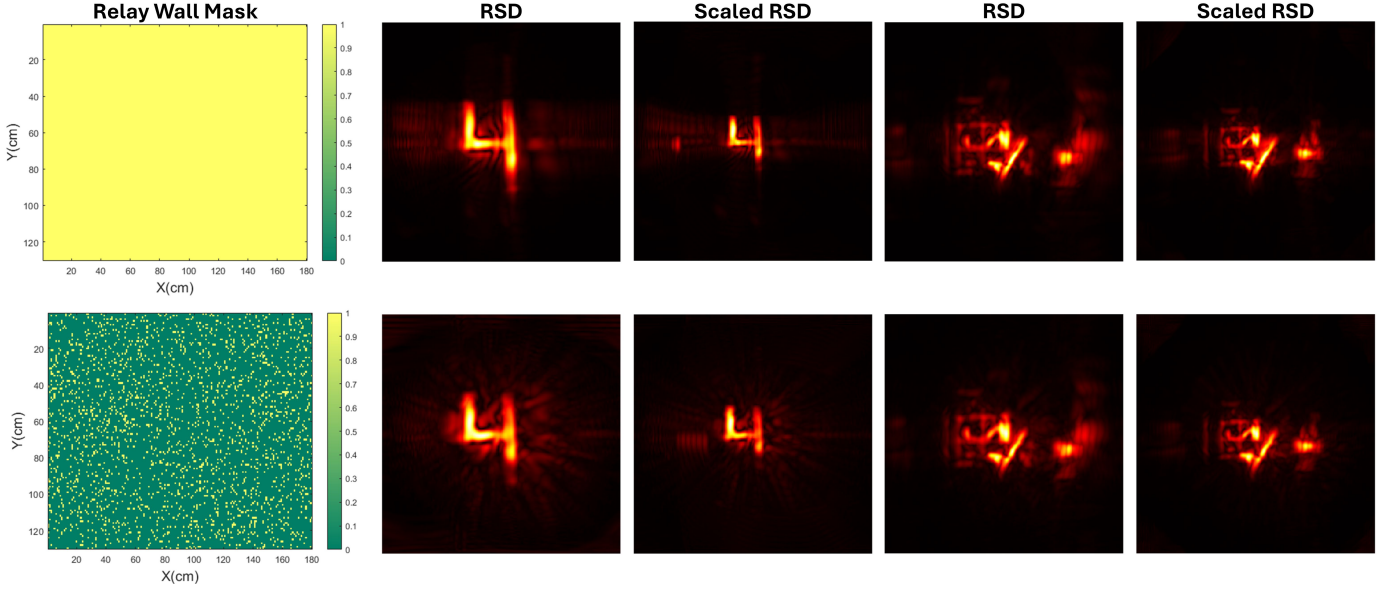


Fig. S3: In Row 1, we show the reconstructions for two different datasets using the Standard RSD (Columns 2 and 4) and Scaled RSD (Columns 3 and 5). In Row 2, we randomly subsample 10% of the relay surface (Column 1) for the same datasets and interpolate to a uniform grid. We show that both Standard RSD (Columns 2 and 4) and the Scaled RSD (Columns 3 and 5) reconstructions do not degrade in quality after discarding 90% of grid positions. Scaled RSD generates reconstructions with perspective projection.

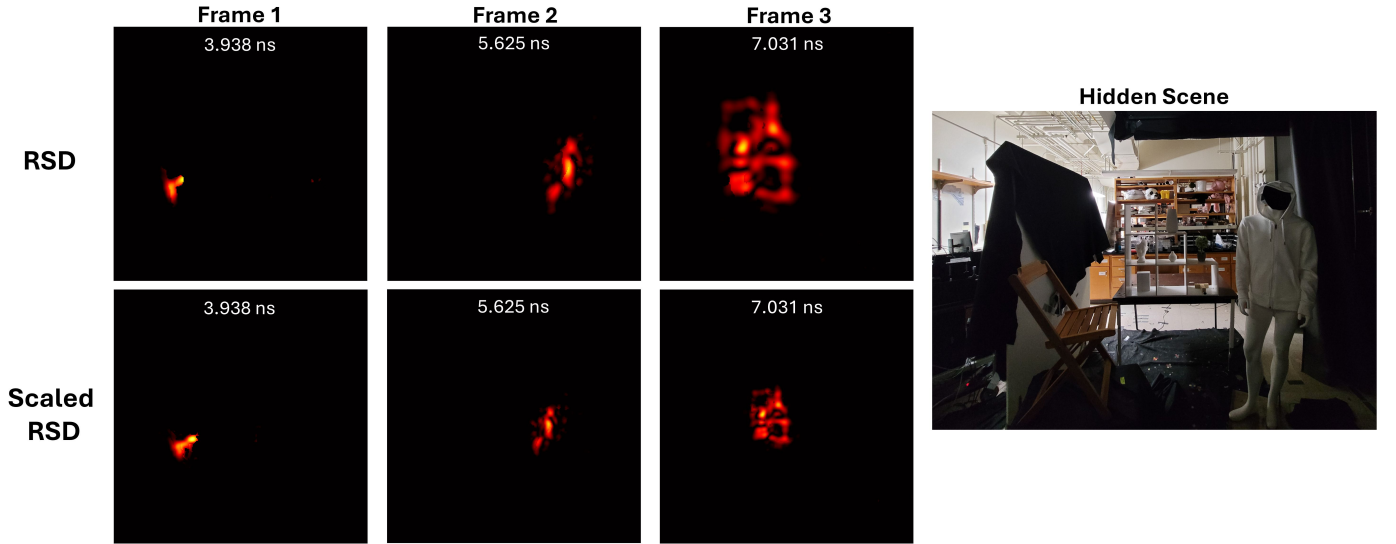


Fig. S4: We generate 4D videos of light transport in the hidden scene using the RSD (Row 1) and the Scaled RSD (Row 2). The mannequin and shelf towards the back of the hidden volume appear smaller in the scaled RSD (Row 2, Columns 3 and 4) due to perspective projection.

S3.3 4D Light Transport

The Standard RSD algorithm can be used to generate a 4D video of our virtual source propagating and interacting with the hidden scene [9]. This algorithm can be adapted to extract complex light transport of the hidden scene [38].

$$I(\vec{x}_v, t) = \iiint \mathcal{P}_{\mathcal{F}}(\vec{x}_p, \omega) e^{-j\frac{\omega}{c}(|\vec{x}_c - \vec{x}_v|)} e^{j\omega t} d\omega d\vec{x}_c d\vec{x}_p \quad (\text{S20})$$

In Row 2 of Fig. S4, we show that the Scaled RSD can be used to generate these 4D videos of the hidden scene, while maintaining perspective projection.

S3.4 Depth-resolved comparison to spherical-grid FBP

To further quantify the relationship between spherical sampling, trapezoidal sampling, and Scaled RSD, we compare reconstructions across the depth sequence shown in Fig. 5. For each depth, spherical-grid FBP is used as the reference. We evaluate two methods against this reference: direct FBP on a trapezoidal grid and Scaled RSD on the same trapezoidal grid. All reconstructions

TABLE S1: Depth-resolved SSIM comparison using spherical-grid FBP as the reference. All methods are evaluated on maximum-intensity projections after maximum-intensity normalization, orientation matching, and 2D correlation alignment. Masked SSIM is computed over pixels above 10% of the spherical FBP reference peak intensity.

Depth (m)	FBP + Trapezoidal		Scaled RSD + Trapezoidal	
	SSIM	Masked SSIM	SSIM	Masked SSIM
1.20	0.913	0.987	0.879	0.977
1.48	0.947	0.995	0.929	0.992
1.80	0.929	0.996	0.927	0.992
2.13	0.929	0.994	0.939	0.995
2.84	0.935	0.996	0.896	0.996
3.32	0.940	0.996	0.849	0.992
3.65	0.913	0.995	0.825	0.990

TABLE S2: Depth-resolved PSNR comparison using spherical-grid FBP as the reference. PSNR is reported in dB. Masked PSNR is computed over pixels above 10% of the spherical FBP reference peak intensity.

Depth (m)	FBP + Trapezoidal		Scaled RSD + Trapezoidal	
	PSNR	Masked PSNR	PSNR	Masked PSNR
1.20	33.02	34.54	30.43	31.61
1.48	34.40	35.46	33.81	35.17
1.80	34.94	37.01	34.39	35.91
2.13	33.92	35.58	34.18	35.61
2.84	34.78	36.32	34.14	37.98
3.32	36.69	39.21	32.30	37.00
3.65	35.79	38.71	31.52	37.46

are converted to 2D maximum-intensity projections, normalized by their maximum intensity, orientation matched, and aligned using 2D correlation before computing SSIM and PSNR.

The trapezoidal-grid FBP reconstructions remain close to the spherical-grid FBP reference across all depths, with SSIM values between 0.913 and 0.947. This supports the use of the trapezoidal grid as a practical approximation to spherical sampling. Scaled RSD also shows strong agreement with the spherical reference at near and intermediate depths, with SSIM values above 0.92 for several depths. At larger depths, the global SSIM decreases, reaching 0.825 at 3.65 m. This decrease is primarily due to background and artifact differences becoming more influential as the projected object occupies fewer pixels. Consistent with this interpretation, masked SSIM computed over the object-support region remains high across all depths, staying above 0.977 for Scaled RSD.

These results show that Scaled RSD preserves the main object structure of the corresponding trapezoidal/spherical reconstructions while avoiding direct spherical-grid evaluation. The comparison also reinforces the intended scope of Scaled RSD: it improves depth-adaptive voxel allocation, memory use, and runtime efficiency, rather than increasing the intrinsic far-range resolution of the NLOS measurement.

S3.5 CPU-Time Scaling under Matched Coverage

We empirically evaluate computational scaling by varying the maximum reconstruction depth from 1.4 m to 10 m while holding the minimum depth at 1.0 m, the depth spacing at 0.04 m, and the near-plane side length at 1.28 m. For each maximum depth, all methods reconstruct matched depth-scaled lateral coverage. Scaled RSD retains 128×128 lateral samples per depth plane, whereas Standard RSD increases the size of its enclosing Cartesian grid to cover the same expanding field of view. At shallow depths, the additional FFT operations make Scaled RSD comparable to or slightly slower than Standard RSD. As the reconstruction depth increases, however, the cost of the enlarged Cartesian grid grows substantially faster. At $z_{\max} = 10$ m, Scaled RSD requires 84.3 s, compared with 1490.6 s for Standard RSD, corresponding to approximately a $17.7\times$ reduction in reported CPU time. Direct depth-scaled FBP is substantially more expensive over the evaluated range and was not evaluated for the larger volumes due to its computational cost. These measurements are consistent with the complexity analysis in Sec. 4.3.

S3.6 Cartesian RSD with Spherical Resampling

We evaluate Cartesian RSD followed by spherical resampling as a post-processing baseline for generating a depth-scaled output. Standard RSD first reconstructs a matched-coverage Cartesian volume containing $350 \times 350 \times 71$ voxels, which is then resampled onto a $190 \times 190 \times 71$ spherical grid. Direct spherical-grid FBP with the same output dimensions is used as the reference. Using the full reconstructed images without foreground masking, Cartesian RSD followed by spherical resampling achieves an SSIM of 0.8573 and a PSNR of 29.84 dB relative to spherical FBP. The comparison shows that post-hoc spherical resampling can approximate the direct spherical reconstruction, but requires first computing and storing the larger Cartesian volume.

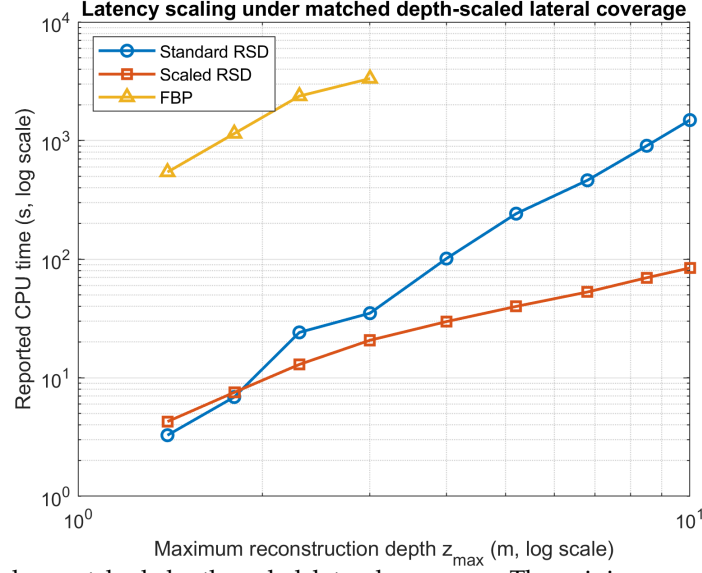


Fig. S5: CPU-time scaling under matched depth-scaled lateral coverage. The minimum reconstruction depth is fixed at 1.0 m, the depth spacing is 0.04 m, and the near-plane side length is 1.28 m. Scaled RSD and depth-scaled FBP use 128×128 lateral samples per depth plane, whereas Standard RSD enlarges its enclosing Cartesian grid to cover the same depth-dependent lateral field of view. All measurements are CPU-only and were obtained on the same workstation. For FBP, we report thread-normalized CPU time using 32 threads. FBP was evaluated only up to $z_{\max} = 3.0$ m because of its substantially higher computational cost. Both axes use logarithmic scales.

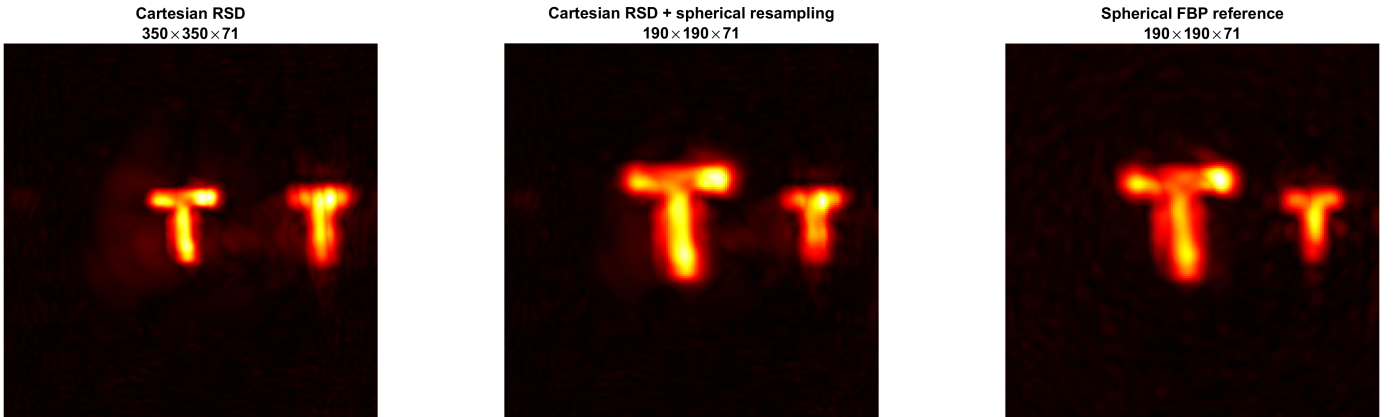


Fig. S6: Comparison of Cartesian RSD, Cartesian RSD followed by spherical resampling, and direct spherical-grid FBP. Standard RSD reconstructs a $350 \times 350 \times 71$ matched-coverage Cartesian volume, which is resampled onto a $190 \times 190 \times 71$ spherical grid. Direct spherical-grid FBP at the same output dimensions is used as the reference. The full-image, unmasked comparison between Cartesian RSD with spherical resampling and spherical FBP gives an SSIM of 0.8573 and a PSNR of 29.84 dB.